8.1.1

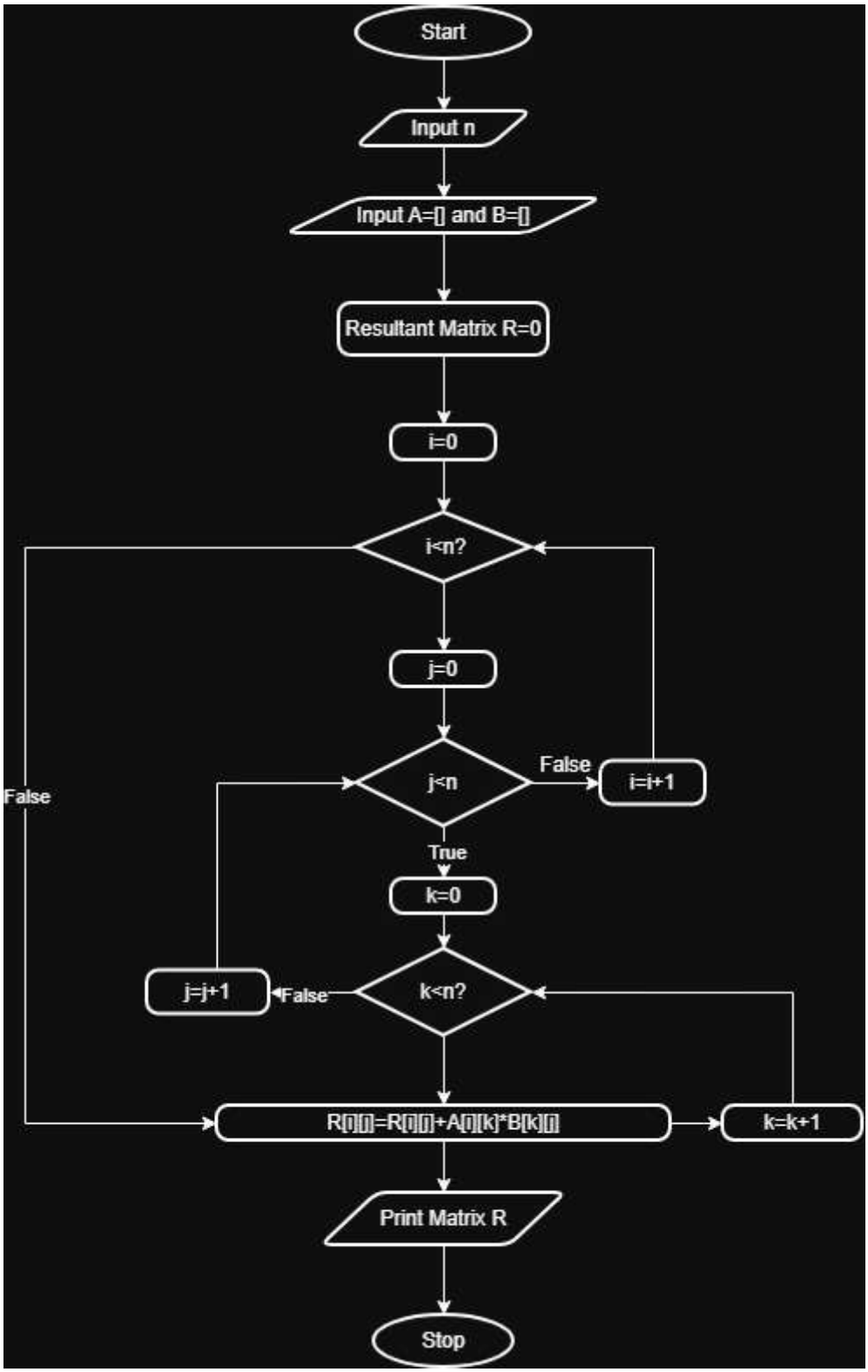
Problem Statement

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

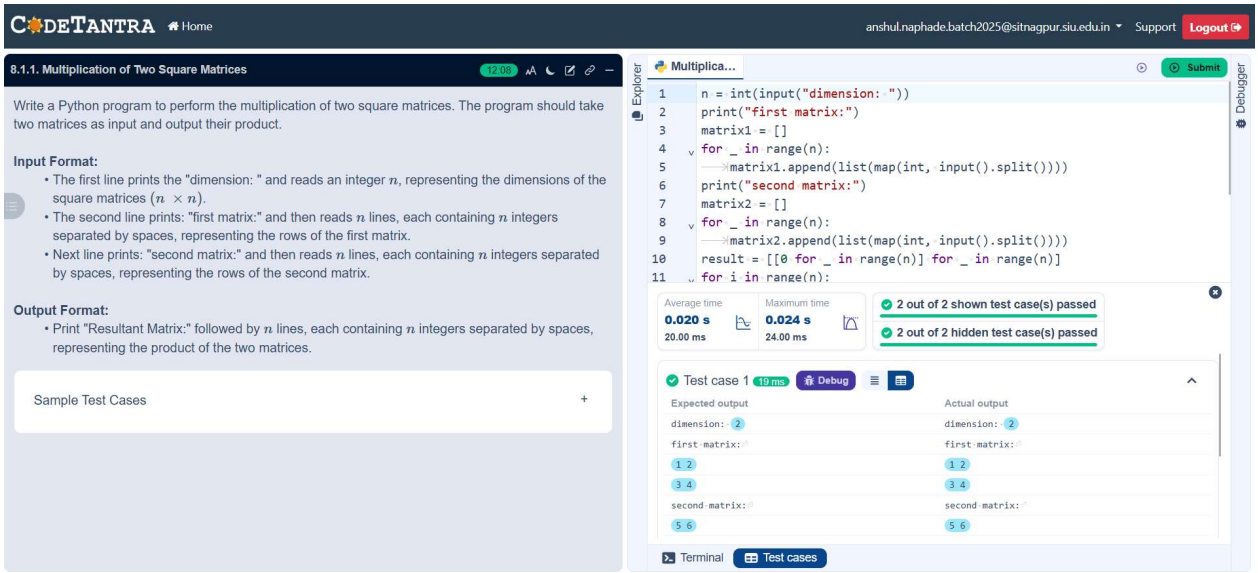
Algorithm

1. Start
2. Input the order *n* of the square matrices
3. Input elements of Matrix *A* and Matrix *B*
4. Create a result matrix *R* of size $n \times n$ and initialize all elements to 0
5. Initialize loop variable *i* from 0 to $n - 1$
6. Inside that, initialize loop variable *j* from 0 to $n - 1$
7. Inside that, initialize loop variable *k* from 0 to $n - 1$
8. Perform multiplication and addition:
   R[i][j]=R[i][j]+A[i][k]×B[k][j]R[i][j] = R[i][j] + A[i][k] \times B[k][j]R[i][j]=R[i][j]+A[i][k]×B[k][j]
9. Repeat until all loops finish
10. Print the result matrix *R*
11. Stop

Flowchart


Start
Input n
Input A=[] and B=[]
Resultant Matrix R=0
i=0
i<n?
j=0
j<n
False
i=i+1
True
k=0
k<n?
j=j+1
False
R[i][j]=R[i][j]+A[i][k]*B[k][j]
k=k+1
Print Matrix R
Stop
False

EXECUTION: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8.1.1. Multiplication of Two Square Matrices
12:08
Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.
Input Format:
• The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices ($n \times n$).
• The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
• Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.
Output Format:
• Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.
Sample Test Cases
Explorer
Multiplica...
Submit
Debugger
n = int(input("dimension: "))
print("first matrix:")
matrix1 = []
for _ in range(n):
matrix1.append(list(map(int, input().split())))
print("second matrix:")
matrix2 = []
for _ in range(n):
matrix2.append(list(map(int, input().split())))
result = [[0 for _ in range(n)] for _ in range(n)]
for i in range(n):
Average time
0.020 s
20.00 ms
Maximum time
0.024 s
24.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
19 ms
Debug
Expected output
dimension: 2
first matrix:
1 2
3 4
second matrix:
5 6
Actual output
dimension: 2
first matrix:
1 2
3 4
second matrix:
5 6
Terminal
Test cases